4 **(a)** **(i)** $V_{out} = 0.040 \times 80 = 3.2\text{V}$ [M1]

potential at A= 0-3.2V=-3.2V[A1]

(ii) $V_x = 8.0 - 3.2 = 4.8V$ [C1]

$\mathrm{R}_x = \dfrac{4.8}{0.040} = 120\Omega$[A1]

(b) **(i)** $R_{out} = \dfrac{100 \times 80}{100 + 80} = 44.44\Omega$ [C1]

$\mathrm{V}_{out} = \dfrac{R_{out}}{R_{total}} \times 8.0$

$= \dfrac{44.44}{44.44 + 120} \times 8.0$ [C1]

= 2.16 V [A1]

(ii) When temperature gets hotter, resistance across the thermistor decreases [M1]

and hence effective/combined resistance across AB decreases. By applying potential divider rule, potential difference across BC increases [M1] which will turn on the fan.

Thus, fan should be connected across BC [A1]